

Exercise 4: Employee Management System

Scenario

You are developing an employee management system for a company. Efficiently managing employee records is crucial.

1. Understand Array Representation

How Arrays are Represented in Memory

An array is a linear data structure in which elements are stored in contiguous memory locations. Each element occupies a fixed amount of memory, and the address of any element can be calculated using its index. This enables direct access to elements in constant time.

Advantages of Arrays

- Stores elements in contiguous memory locations.
- Provides fast access using indices.
- Simple to implement and efficient for fixed-size collections.
- Requires less memory overhead compared to linked data structures.
- Suitable when the number of elements is known in advance.

2. Setup

A Java project named `EmployeeManagementSystem` was created.

The project contains a code folder with the following files:

`EmployeeManagementSystem`

└── code

 └── Employee.java

 └── EmployeeManagementSystem.java

- `Employee.java` – Defines the `Employee` class with attributes such as `employeeId`, `name`, `position`, and `salary`.
- `EmployeeManagementSystem.java` – Uses an array to store employee records and implements `add`, `search`, `traverse`, and `delete` operations.

3. Implementation

Employee Class

The Employee class contains the following attributes:

- employeeId
- name
- position
- salary

Employee Management Operations

The employee records are stored using an array of Employee objects.

The following operations are implemented:

- Add Employee
- Search Employee
- Traverse Employees
- Delete Employee

4. Analysis

Time Complexity

Operation	Time Complexity
Add Employee	$O(1)$
Search Employee	$O(n)$
Traverse Employees	$O(n)$
Delete Employee	$O(n)$

Limitations of Arrays

- Fixed size; the array cannot grow dynamically once created.
- Insertion and deletion may require shifting elements.
- Memory may be wasted if the array is not fully utilized.
- Searching is slower compared to hash-based data structures.

When to Use Arrays

Arrays are suitable when:

- The number of records is fixed or changes infrequently.

- Fast index-based access is required.
- Memory efficiency is important.
- The application performs more read operations than insertions or deletions.

5. Conclusion

The Employee Management System was successfully implemented using an array to store employee records. The system supports adding, searching, traversing, and deleting employees efficiently. Arrays provide fast direct access to records but have limitations such as fixed size and slower insertion and deletion operations. They are best suited for applications where the number of records is relatively stable.